CEN 107 MATHEMATICS I FINAL EXAMINATION **20 January 2021**

Full Name:...

Student Number:... **Signature:** ....

Fill in the blanks in question 1.

1. Let us consider the rational function $f(x) = \frac{x}{x^2-1}$ for which domain is .________________and the root is .________

$\lim_{x\to(-1)^-} \frac{x}{x^2-1} =$.______ and $\lim_{x\to1^+} \frac{x}{x^2-1} =$.______ , hence $x = \pm1$ are .____________________ asymptotes,

$\lim_{x\to-\infty} \frac{x}{x^2-1} =$.______ and $\lim_{x\to\infty} \frac{x}{x^2-1} =$.______ , hence __________ is a ______________________ asymptote.

$f'(x) = \frac{\dots\dots\dots\dots}{(x^2-1)^2}$ and thus no critical values for relative max or min points. The second derivative is

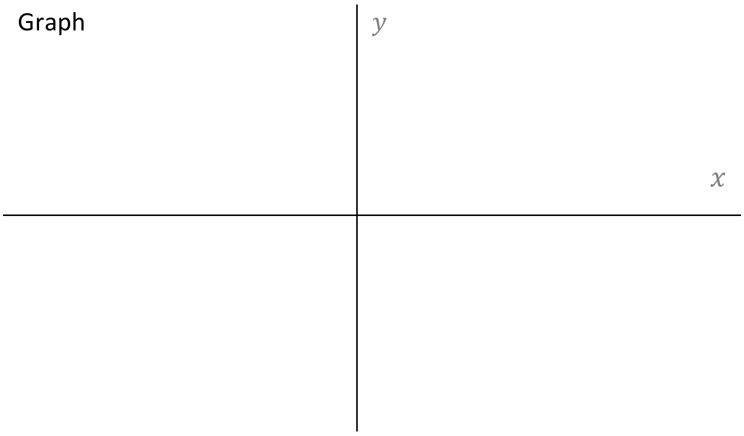
$f''(x) = \frac{2x(x^2+3)}{\dots\dots\dots\dots}$ and hence the only critical value for inflection point is ____________ .

Table -1 0 1

f'				
f''				
f				

inflection point: $(\dots , \dots)$

Graph



2. Compute the following derivatives.

$\left(\ln(1+e^{2x})\right)'$ $\left(\arcsin(x^5+1)\right)'$ $((\arctan x)^3)'$

$\left(\cos(\mathrm{Arctan}(2x))\right)'$ $\left(\sqrt[3]{\mathrm{Arctan}(\sec x)}\right)'$ $\left(\mathrm{Arctan}(\sin x)\right)'$

3. Calculate the following limits. **(Do not use L'Hospital's Rule)**

$\lim_{x\to-\infty} \left(x+\sqrt{x^2+3x}\right)$ $\lim_{x\to1^-} \frac{3|x-1|+[\![x+1]\!]+5x}{\sqrt{2}x} =$

$\lim_{x\to-\infty} \frac{\sqrt{x^2+x+1}}{x-5}$ $\lim_{x\to0} \frac{\tan(\sqrt{3}x)}{7x} =$